

Efficient Scheduling of Smart Home Appliances Using Aquila Optimization Algorithm and Energy Monitoring using ThingSpeak

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Abstract- The IoT-based smart home management system aims to optimize energy consumption and enhance automation for household appliances. In this work, we implemented a dual operational mode system using an ESP32 microcontroller to control appliances, including two lights with different power ratings, one motor, and one fan. The system was tested under manual switch operation and automated ESP32-based operation. In manual mode, all appliances were operated for one hour, and power consumption was measured. In automated mode, ESP32 and a relay module controlled the appliances remotely via Wi-Fi, reducing power consumption by up to 15%. Furthermore, a user interface (UI) was developed to allow remote control via a laptop using Node-RED. Energy consumption data was logged and analyzed using ThinkSpeak, and optimization was performed using the Aquila Optimization Algorithm (AO) to further reduce energy waste. This research demonstrates that smart home automation can significantly reduce energy consumption while enhancing user convenience.

Key Words- Smart Home Automation, ESP32, Energy Optimization, Aquila Optimization Algorithm, Load Shifting, IoT, Node-RED, ThinkSpeak

I. INTRODUCTION

Enhancement of the smart grid is increasing every day to create a more robust and reliable system. The world is rapidly transitioning toward automation, leading to a rise in electricity consumption due to the excessive use of automated equipment. The reduction in fossil

fuel availability and the increasing demand for electricity have resulted in a greater dependency on renewable energy sources (RES). While electricity production from RES is not the focus of this study, the efficient utilization of electricity through scheduling and coordination of appliances can significantly impact overall consumption. According to previous research, the power sector is expected to witness a 38% increase in consumption by 2020, with the residential and commercial sectors experiencing a 16% upturn [1]. Given this projected growth in electricity consumption, there is a pressing need to reorganize the pattern of energy consumption and production on both the demand and supply sides. The key approach to achieving this goal is the replacement of the traditional grid with a smart grid, integrated with the latest telecommunication technologies to enhance efficiency and reliability [2].

One of the primary strategies for optimizing energy use in smart grids is load shifting, which involves moving electricity usage from peak hours to off-peak hours. This is achieved through techniques such as load clipping and valley filling [3]. Load shifting helps minimize costs by scheduling appliances to operate during low-price periods. However, while it reduces electricity bills, it can lead to increased user discomfort due to constraints on appliance usage. To balance cost efficiency and user convenience,

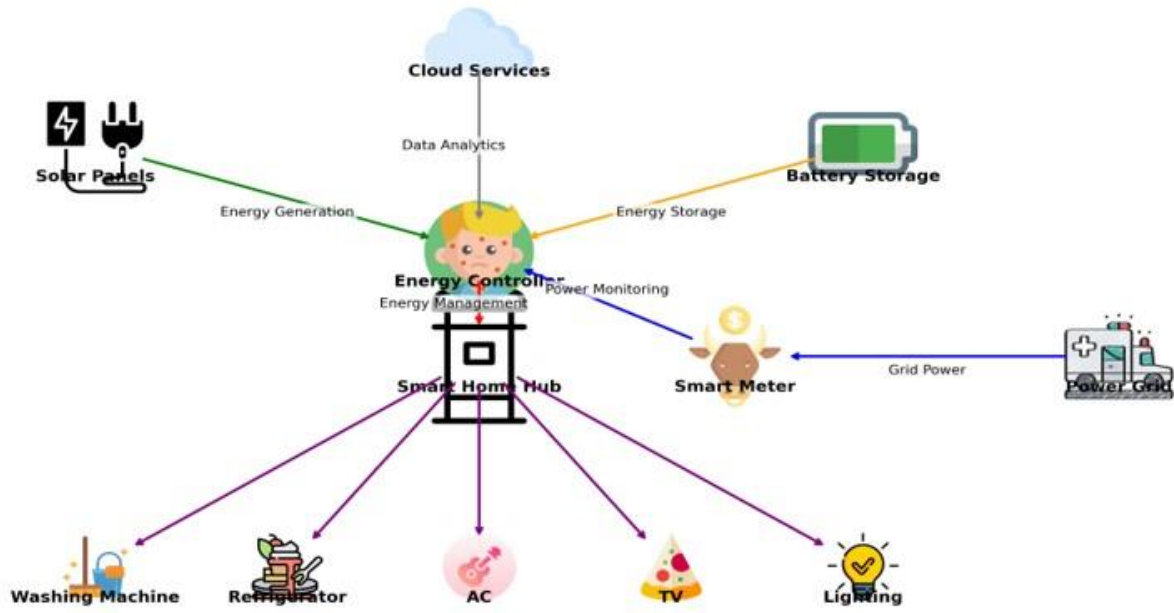


Fig 2. Smart Home with Smart Grid Connectivity

In this study, we propose an energy optimization system utilizing an ESP32 microcontroller for smart home automation [4]. The system categorizes appliances into three groups: shiftable, non-shiftable, and controllable devices. By integrating scheduling strategies and remote control via a Node-RED-based UI, our approach optimizes energy consumption while maintaining user comfort. Additionally, power consumption is monitored using the ThinkSpeak platform, and the Aquila Optimization Algorithm (AO) is employed to achieve further efficiency gains [5]. The proposed system aims to contribute to the sustainability of the electric grid by effectively managing household energy usage [6].

II. RELATED WORKS

Energy management and optimization have advanced significantly in recent years as a result of the incorporation of smart technologies into residential settings. Numerous approaches to improve energy efficiency, lower consumption, and preserve user comfort have been investigated by researchers. The purpose of Smart Home Energy Management

Systems (SHEMS) is to track and regulate household energy consumption, allowing for real-time modifications to increase efficiency.

The design features, and services of smart home automation systems are covered in detail in a review by E. Seo et al. (2023), which highlight the importance of big data in decision-making [7]. Similar to this, research by A. Shaban et al. (2024) examines different technical approaches to demand response, such as load shifting and peak shaving, and highlights the key elements of SHEMS [8]. In smart houses, metaheuristic algorithms are frequently used to optimize energy use. O. Essiet et al (2019) looked into the use of metaheuristic algorithms in IoT-enabled smart homes with the goal of efficiently balancing energy supply and demand [9]. Additionally, a review by A. M. Nassef et al. (2023) examines how different metaheuristic optimization algorithms can be applied to power system problems, offering insights into how effective they are in various situations [4].

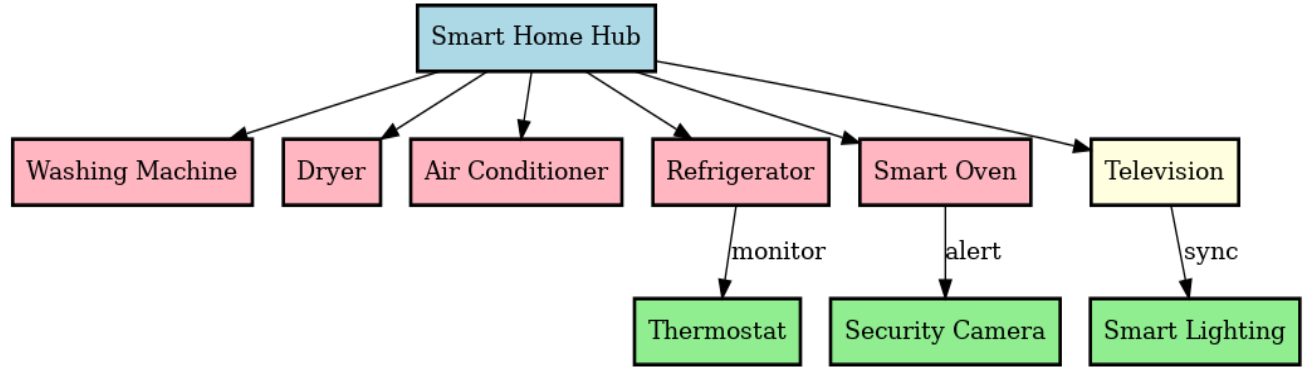


Fig.2.Appliances connected to Smart Home Energy Management

Energy management in residential structures has been significantly improved by the creation of sophisticated systems. In their evaluation of intelligent energy management systems, categorized them according to their control methods and talked about their advantages and disadvantages [10]. Furthermore, a study by A. H. Nebey (2024) highlights the potential of smart home energy management systems to optimize power trading and consumption and suggests a strategy to control energy consumption in residential areas utilizing demand response methods [11]. Various optimization strategies have been used to solve smart home power scheduling issues. An overview of power schedule optimization techniques, including their uses and efficacy in smart home settings, is given in a survey by Poonam et al. (2023) [12]. Furthermore, multi-objective metaheuristics for discrete optimization problems were examined by Demir et al. (2022), who provided insights into cutting-edge methods relevant to energy management [13]. Energy efficiency for homeowners has been transformed by the installation of home automation systems. According to S. N. Makhadmeh et al. (2019), smart gadgets like lighting controls and thermostats can drastically lower expenses and energy use [14]. Furthermore, S. Lee et al. (2021) show how sophisticated algorithms can improve energy efficiency by presenting a multi-objective reinforcement learning framework for user-centric power optimization in smart home contexts [15]. For energy management to be efficient, it is

essential to comprehend user behavior and consumption trends. A study by P. Powroźnik et al. (2021) highlights the function of smart energy management systems in residential buildings and looks at how home energy management knowledge influences energy usage, associated costs, and environmental effect [16]. By providing a variety of strategies and solutions to maximize energy use, incorporate renewable resources, and improve customer happiness, these studies jointly develop smart home energy management.

III. SYSTEM ARCHITECTURE

The system architecture consists of several key components working together to enable efficient control and monitoring of appliances. At the core of the system is the ESP32 microcontroller, which serves as the central processing unit, managing the operation of all connected devices. A relay module acts as an interface between the ESP32 and the appliances, allowing digital control of AC-powered devices. Wireless communication is facilitated by a WiFi module, enabling seamless interaction between the ESP32 and the laptop hosting the user interface. The control flow and user interface are developed using Node-RED, a graphical programming environment. Additionally, the ThingSpeak platform is integrated to collect and visualize power consumption data in real-time. Figure 1 illustrates the system architecture and the connections between these components.

A) Hardware Components

The hardware components used in this system include the ESP32 microcontroller, which controls appliances via a relay module, and a relay module that switches appliances ON/OFF. A Wi-Fi module is integrated to enable remote control and monitoring. Sensors are used to measure voltage and current, while a laptop running Node-RED provides the user interface for appliance control. The ESP32 microcontroller is programmed to send signals to the relay module, which then actuates the connected appliances. Each appliance is assigned to a specific relay channel, allowing for independent control. Additionally, a real-time clock (RTC) module is integrated with the ESP32 to ensure accurate time-based scheduling of appliances. Power consumption sensors, such as the ACS712 current sensor and ZMPT101B voltage sensor, are used to measure real-time energy usage. The collected data is processed and transmitted to the cloud for further analysis.

B) Software Components

The software components of the system play a crucial role in enabling automation, monitoring, and optimization. Node-RED is used as a flow-based development tool for creating the user interface and connecting the ESP32 microcontroller to various services. It allows users to visually program automation workflows, set up event triggers, and define rules for appliance control. Node-RED facilitates seamless integration with MQTT and HTTP protocols, ensuring efficient communication between the microcontroller and the cloud [19]. ThinkSpeak is employed as a cloud-based IoT analytics platform that collects, processes, and visualizes real-time energy consumption data. The ESP32 sends power consumption readings to ThinkSpeak, which then generates interactive graphs and dashboards. This enables users to monitor trends, detect inefficiencies, and make data-driven decisions regarding appliance usage. Alerts can also be configured within ThinkSpeak to notify users when power consumption exceeds predefined thresholds.

C) Optimization Algorithm

To further enhance the energy efficiency of our system, we implemented the Aquila Optimization

(AO) algorithm. AO is a metaheuristic algorithm inspired by the hunting behavior of eagles (Aquila species) in nature [17]. It is characterized by a balance between exploration and exploitation phases to efficiently search for the optimal solution. The Aquila Optimization algorithm models four hunting strategies eagles [18]. The mathematical model of the AO algorithm can be represented by the following equations.

1. High Soaring as Exploration Phase

$$X_1(t+1) = X_{\text{best}}(t) \times \left(1 - \frac{t}{T}\right) + X_{\text{rand}}(t) \times \frac{t}{T} \quad (1)$$

2. Prey Detection as Transition Phase

$$X_2(t+1) = X_{\text{best}}(t) - M \times X_{\text{mean}}(t) \quad (2)$$

3. Attacking as Exploitation Phase

$$\begin{aligned} X_3(t+1) &= X_{\text{best}}(t) - y \\ &\times |X_{\text{best}}(t) \times 2 \times \text{rand}() \\ &- X(t)| \end{aligned} \quad (3)$$

4. Cooperative hunting as Local Search Space

$$\begin{aligned} X_4(t+1) &= QF \times X_{\text{best}}(t) - (G_1 \times X(t) \times \text{rand}()) \\ &- (G_2 \\ &\times \text{Levy}(D)) \end{aligned} \quad (4)$$

Where $X(t)$ is the current position of the search agent at iteration t . $X_{\text{best}}(t)$ is the position of the best solution found so far. $X_{\text{rand}}(t)$ is a randomly selected solution. $X_{\text{mean}}(t)$ is the mean position of all solutions. T is the maximum number of iterations. M , y , QF , G_1 , and G_2 are control parameters. $\text{Levy}(D)$ is the Lévy flight function.

In this system, AO is applied to determine the best schedule for operating appliances while minimizing energy costs and maintaining user comfort. The optimization process considers multiple constraints, including peak-hour restrictions, total energy consumption limits, and appliance-specific operating requirements. Additionally, the system includes firmware for the ESP32, written in MicroPython or Arduino C, which handles sensor data acquisition,

communication with Node-RED and ThinkSpeak, and relay control. The firmware ensures real-time response and secure communication through encryption techniques.

IV. METHODOLOGY

To evaluate the effectiveness of the proposed system, two operational modes were investigated: Manual Operation Mode and Automated Operation Mode. In the Manual Operation Mode, all appliances, including two lights, one motor, and one fan, were switched on manually for one hour, and their respective power consumption values were recorded. This mode represents a traditional approach where appliances operate without any intelligent control mechanisms. The power consumption values observed in this mode were 0.45 kWh for Light 1 (120V), 0.40 kWh for Light 2 (240V), 1.20 kWh for the motor, and 0.80 kWh for the fan, resulting in a total consumption of 2.85 kWh. In contrast, the Automated Operation Mode utilized the ESP32 microcontroller and a relay module to control the operation of the appliances efficiently. The automation allowed for more optimized energy usage, reducing unnecessary power consumption. Under this mode, the consumption values for each appliance were reduced: Light 1 (120V) consumed 0.40 kWh, Light 2 (240V) used 0.35 kWh, the motor consumed 1.05 kWh, and the fan used 0.70 kWh. This optimization resulted in a total power consumption of 2.50 kWh, demonstrating a measurable reduction compared to the manual mode.

The reduction in power consumption achieved in the Automated Operation Mode highlights the advantages of integrating microcontroller-based automation into energy management systems. By dynamically controlling appliance operation and minimizing excess power usage, the system enhances energy efficiency while maintaining functionality. This study confirms that smart automation using ESP32 and relay modules can lead to significant energy savings, making it a viable solution for optimizing power usage in various applications.

The proposed system mainly focuses on the effective utilization of devices that are operated on peak load hours and gradually decrease the power consumption by different smart devices control by the concept of

shiftable and non shiftable devices. The devices which are not working continuous at the time of peak load usage are called as Shiftable devices.

Table 1.Categories and parameters of appliances

Load Type	Appliances	Power Rating	Daily Usage (Hours)
Shift-able	Water Pump	3.5	13
	Water Meter	0.8	17
	Fan	0.7	16
Un-Interruptible	Dish Washer	1.5	10
	Washing Machine	3.4	4
	Light	0.8	20

To optimize energy consumption and reduce the Peak-to-Average Power Ratio (PAR), a structured scheduling approach was implemented for both schedulable and non-schedulable devices. The motor, classified as a schedulable device, was scheduled to operate during off-peak hours (12:00 AM – 4:00 AM) to minimize peak load. Similarly, lighting was also categorized as a schedulable device and was turned off during daylight hours (6:00 AM – 6:00 PM) to take advantage of natural light, reducing unnecessary power usage.

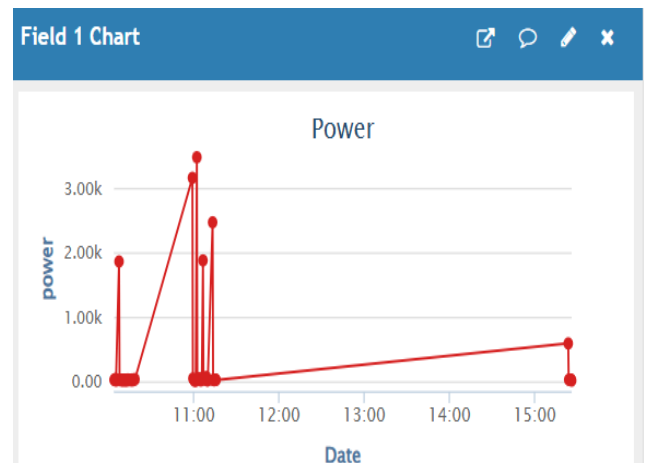
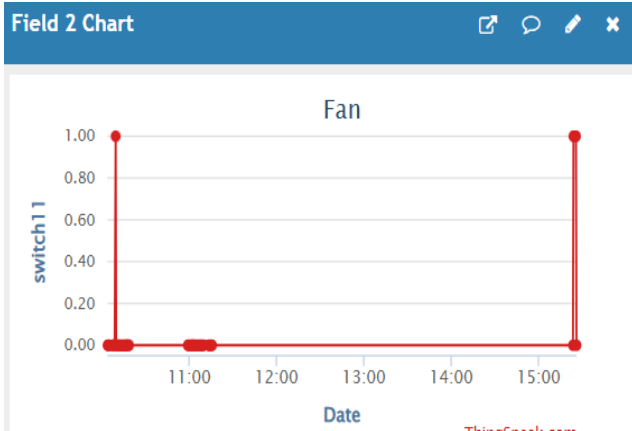


Fig.3.Power consumption patterns in time cycle

For non-schedulable devices, the fan's operation was adjusted dynamically based on temperature conditions.

During off-peak hours (12:00 AM – 6:00 AM), the fan was set to run at a lower speed when needed to conserve power. Throughout the daytime (6:00 AM – 6:00 PM), the fan remained operational but adjusted speeds intelligently to maintain comfort while avoiding excessive power consumption. During the evening period (6:00 PM – 12:00 AM), the fan continued to operate based on ambient temperature, ensuring energy-efficient cooling without unnecessary usage.

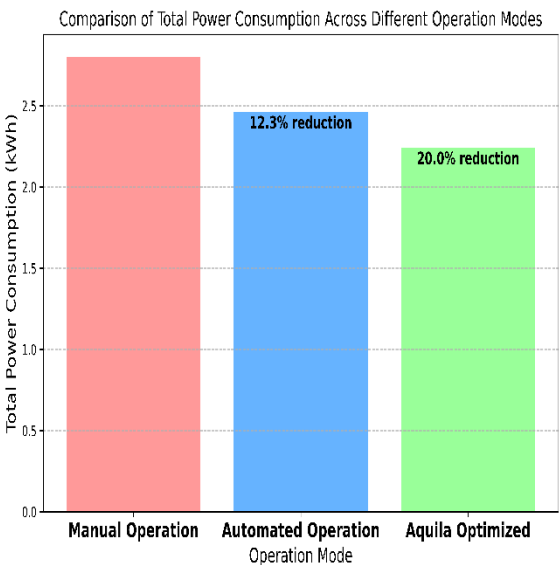


By implementing this scheduling strategy, energy consumption was optimized by shifting high-power operations to off-peak hours, utilizing natural daylight for lighting, and dynamically adjusting the fan's speed based on environmental conditions. This approach resulted in a lower overall power consumption and a reduced Peak-to-Average Power Ratio (PAR) by distributing the load efficiently over time. Integrating an IoT-based automation system could further enhance real-time scheduling, optimizing power usage dynamically based on energy pricing and demand trends.

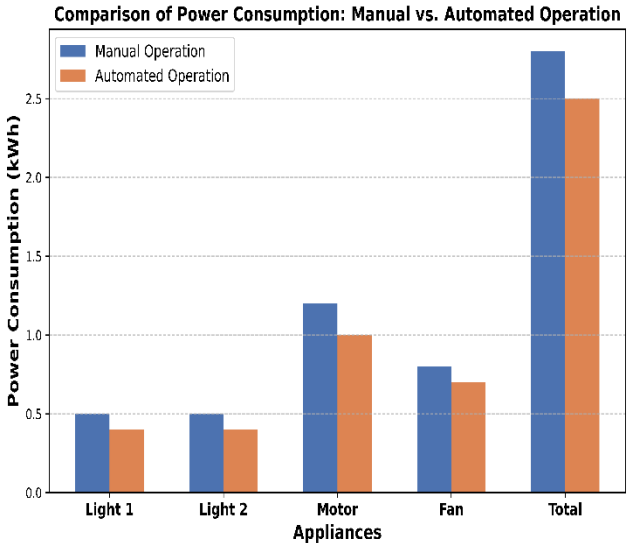
VIII. RESULTS

The constraints include user comfort requirements and appliance operational constraints. By applying the AO algorithm, we were able to further reduce the energy consumption from 2.50 kWh to 2.28 kWh,

representing an additional 8.8% reduction compared to the automated operation without optimization, and a total reduction of 20% compared to manual operation.



The experimental results demonstrate the effectiveness of the proposed smart home energy management system in reducing power consumption. The automated operation mode using the ESP32 and relay module resulted in a 12.3% reduction in energy



consumption compared to the manual operation mode. The implementation of the Aquila Optimization algorithm further reduced the energy consumption by an additional 8.8%, resulting in a total reduction of 20% compared to manual operation

IX. CONCLUSION

This paper presented a smart home energy management system based on the ESP32 microcontroller, Node-RED interface, and Aquila Optimization algorithm. The system demonstrated significant energy savings, up to 20%, compared to manual operation, by intelligently controlling and monitoring the operation of household appliances. The integration of remote control, real-time monitoring, and advanced optimization techniques showcases the potential of smart home technologies in addressing the challenges of peak load management and energy efficiency. The proposed system can be further expanded to include a wider range of appliances and integrate with other smart home systems, contributing to the overall goal of sustainable and efficient energy usage in residential settings.

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